LAB 5: CONFIGURATION OF STATIC ROUTES AND DEFAULT ROUTES

Objectives

1. To understand the concept of Static Routing and Default Routing for manual path selection in a network.
2. To configure a network topology using Cisco Packet Tracer that connects different networks using Static and Default routes to visualise the flow of packets between different networks.

Software and hardware used:

1. Cisco Packet Tracer

2. PC

Theory

1. Static Routing

Static route implementation refers to the manual process of creating fixed routing paths within a router's forwarding database. Network administrators explicitly specify the exact route that network traffic should traverse to reach a particular destination subnet.

Functionality: The configuration is accomplished through the command

```
ip route [destination_network] [subnet_mask] [next_hop_address]
```

When packets arrive, the router consults these predefined entries to make forwarding decisions. If connectivity is disrupted, administrators must manually intervene to reconfigure or remove the affected routing entry.

Use Cases: Used in security objectives by preventing automatic route advertisement, effectively concealing network segments from potential threats.

2. Default Routing

Default routing is a specific type of static routing that acts as a "gateway of last resort." It allows the router to forward all packets to a specific next-hop router if the destination network is not found in the routing table.

Operation: It involves configuring a route with all zeros:

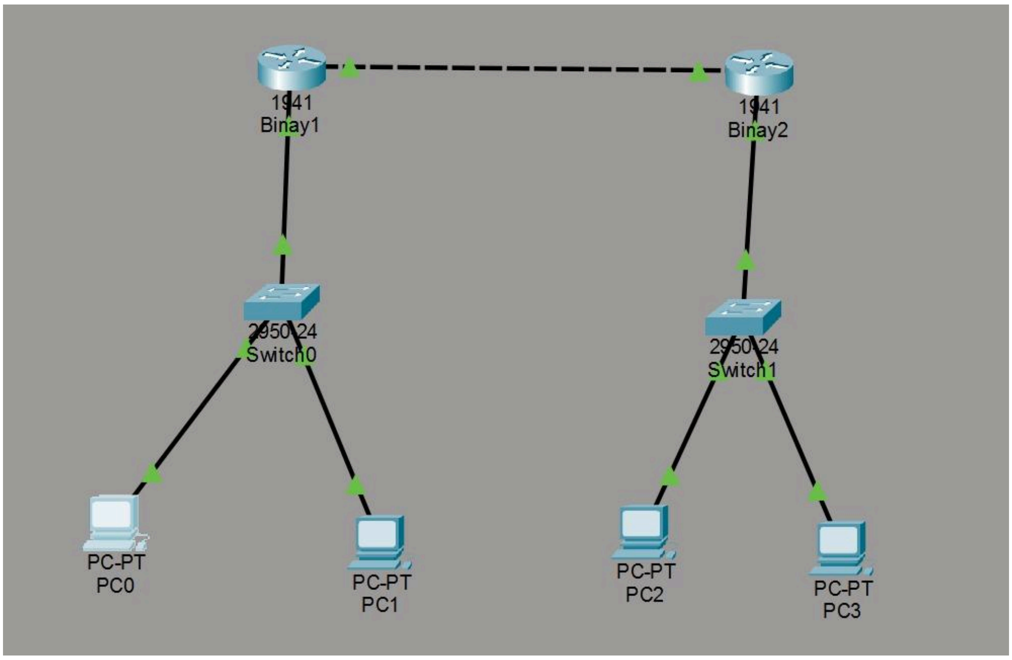
```
ip route 0.0.0.0 0.0.0.0 [next_hop_address]
```

This tells the router to send any unknown traffic to the specified neighbor.

Use Cases: Used primarily in stub networks (networks with only one exit path) or to connect a network to the Internet via an ISP. It reduces the size of the routing table.

Network Topology:

The network infrastructure consists of two routing devices identified as Binay1 and Binay2, interconnected through a serial communication link. Within the first segment, switch0 provides network access for workstations PC0 and PC1, connecting them to the Binay1 router. The second segment features Switch1, which enables connectivity for workstations PC2 and PC3 to Binay2. This configuration represents a basic multi-segment network requiring proper routing configuration for inter network communication.



Configuration Table:

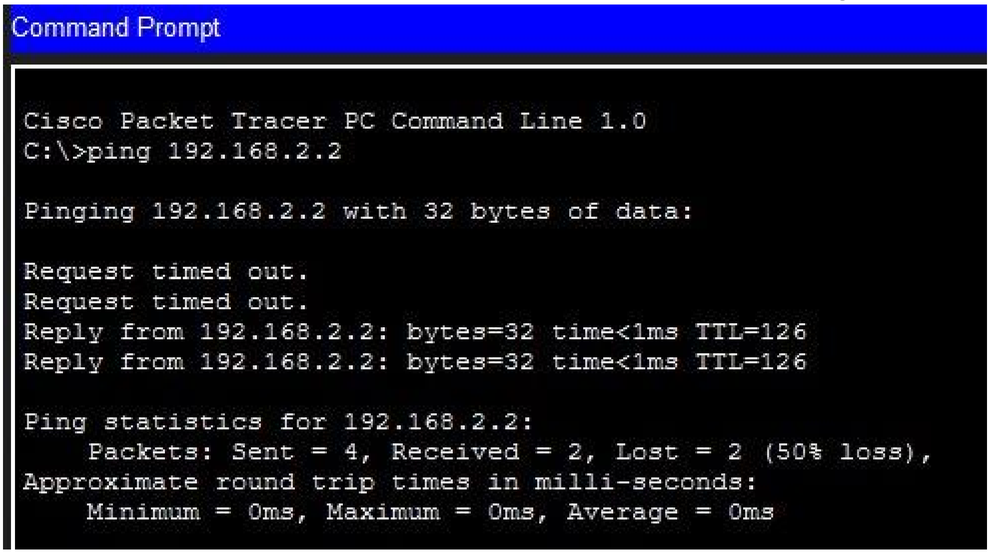
Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Binay1	GigabitEthernet0/0	10.0.0.1	255.0.0.0	N/A
Binay1	GigabitEthernet0/1	192.168.1.1	255.255.255.0	N/A
Binay2	GigabitEthernet0/0	10.0.0.2	255.0.0.0	N/A
Binay2	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

Routing Configuration

Device	Routing Type	Command / Route
Binay1	Static Route	ip route 192.168.2.0 255.255.255.0 10.0.0.2
Binay2	Static Route	ip route 192.168.1.0 255.255.255.0 10.0.0.1
Binay1	Default Route (Alt)	ip route 0.0.0.0 0.0.0.0 10.0.0.2

Result:

The network was successfully configured. To verify the connectivity, a ping request was sent from PC1 (192.168.1.3) in the first network to PC2 (192.168.2.3) in the second network. The packets were sent and received successfully which validates that the static route is correct and similar was done for default route connection testing.



Discussion and Conclusion:

During this Lab we successfully implemented static and default routing using Cisco Packet Tracer. Static routing provided precise control over traffic paths and enhanced security, while default routing simplified the routing table for stub networks. The hands-on implementation demonstrated how routers forward packets based on implementation demonstrated how routers forward packets based on manually configured entries. The lab was completed successfully, providing essential knowledge of routing principles for network administration.